

Fundamental Math/Physics in Computer Graphics (I)

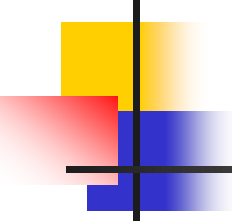
Lecture 3

Dan Biediger



UNIVERSITY *of* **HOUSTON**

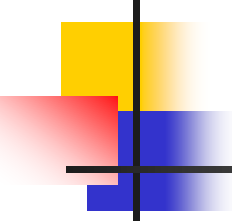
DEPARTMENT OF COMPUTER SCIENCE



What is Computer Graphics?

Using a computer to convert a model into an image



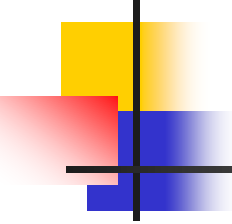


3D Geometry

Need a way to represent the model

- Objects
 - Positions
 - Geometries
 - Properties
- Cameras
 - Position and Orientation
 - Properties

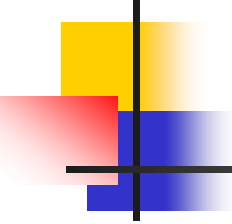




Basic Types

- Scalars: number $s = 3$
- Points: vector $p = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$
- Directions: vector $d = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$



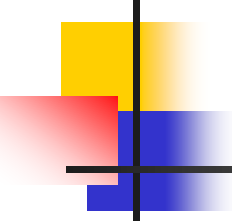


Transformations

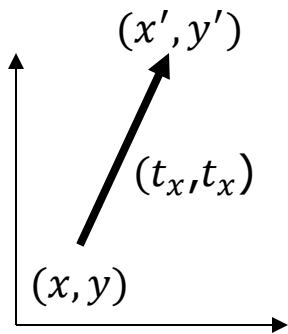
Why use transformations?

- Create object in convenient coordinates
- Reuse basic shape multiple times
- Hierarchical modeling
- System independent
- Virtual cameras





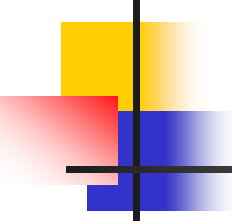
Translation



$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \\ t_z \end{bmatrix}$$

$$T(t_x, t_y, t_z) \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \\ t_z \end{bmatrix}$$





Properties of Translation

$$T(0,0,0) \mathbf{v} = \mathbf{v}$$

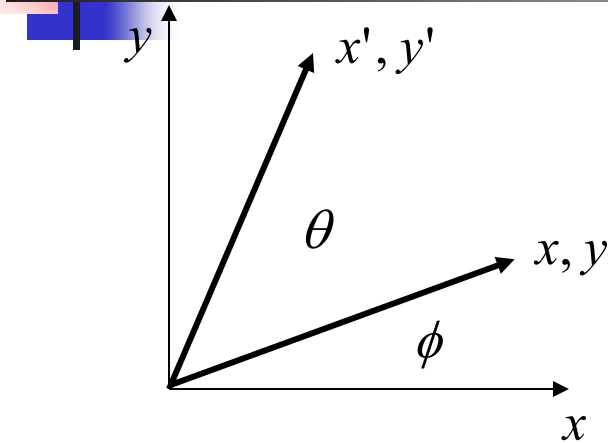
$$T(s_x, s_y, s_z) T(t_x, t_y, t_z) \mathbf{v} = T(s_x + t_x, s_y + t_y, s_z + t_z) \mathbf{v}$$

$$T(s_x, s_y, s_z) T(t_x, t_y, t_z) \mathbf{v} = T(t_x, t_y, t_z) T(s_x, s_y, s_z) \mathbf{v}$$

$$T^{-1}(t_x, t_y, t_z) \mathbf{v} = T(-t_x, -t_y, -t_z) \mathbf{v}$$



Rotations (2D)



$$\begin{aligned}x' &= x \cos \theta - y \sin \theta \\y' &= x \sin \theta + y \cos \theta\end{aligned}$$

$$x = r \cos \phi$$

$$y = r \sin \phi$$

$$x' = r \cos(\phi + \theta)$$

$$y' = r \sin(\phi + \theta)$$

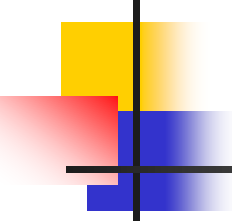
$$\cos(\phi + \theta) = \cos \phi \cos \theta - \sin \phi \sin \theta$$

$$\sin(\phi + \theta) = \cos \phi \sin \theta + \sin \phi \cos \theta$$

$$x' = (r \cos \phi) \cos \theta - (r \sin \phi) \sin \theta$$

$$y' = (r \cos \phi) \sin \theta + (r \sin \phi) \cos \theta$$





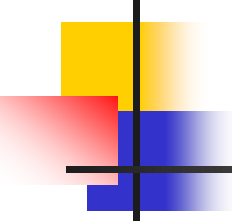
Rotations (3D)

$$R_x(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix}$$

$$R_y(\theta) = \begin{bmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{bmatrix}$$

$$R_z(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$





A Note on Rotations

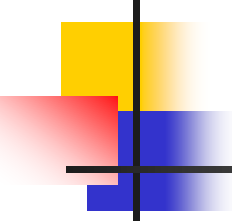
$$Rx(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix}$$

$$\det(Rx(\theta)) = \begin{vmatrix} + & + & + & - & - & - \\ 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta & 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta & 0 & \sin \theta & \cos \theta \end{vmatrix}$$


$$\det(Rx(\theta)) = 1 \cdot \cos \theta \cdot \cos \theta + 0 \cdot -\sin \theta \cdot 0 + 0 \cdot 0 \cdot \sin \theta \\ - 1 \cdot -\sin \theta \cdot \sin \theta - 0 \cdot 0 \cdot \cos \theta - 0 \cdot \cos \theta \cdot 0$$

$$\det(Rx(\theta)) = \cos^2 \theta + \sin^2 \theta = 1$$





Properties of Rotations

$$R_a(0) = I$$

$$R_a(\theta)R_a(\phi) = R_a(\phi + \theta)$$

$$R_a(\theta)R_a(\phi) = R_a(\phi)R_a(\theta)$$

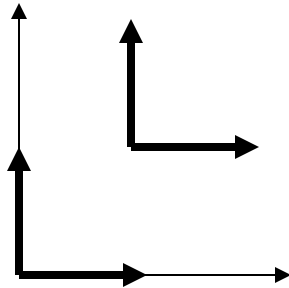
$$R_a^{-1}(\theta) = R_a(-\theta) = R_a^T(\theta)$$

$$R_a(\theta)R_b(\phi) \neq R_b(\phi)R_a(\theta) \quad \text{order matters!}$$

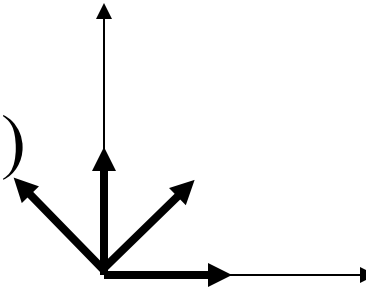


Combining Translation & Rotation

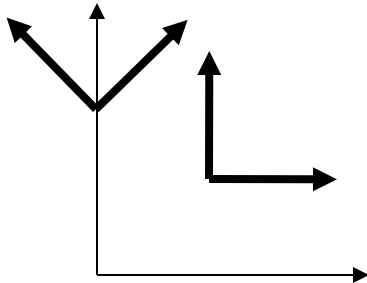
$T(1,1)$



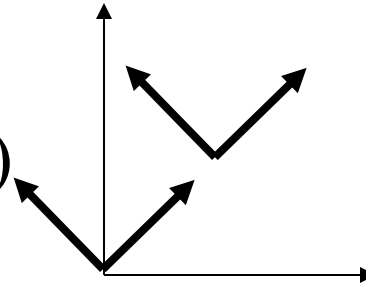
$R(45^\circ)$



$R(45^\circ)$



$T(1,1)$



Combining Translation & Rotation

$$\mathbf{v}' = \mathbf{v} + T$$

$$\mathbf{v}'' = R\mathbf{v}'$$

$$\mathbf{v}'' = R(\mathbf{v} + T)$$

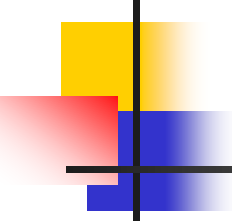
$$\mathbf{v}'' = R\mathbf{v} + RT$$

$$\mathbf{v}' = R\mathbf{v}$$

$$\mathbf{v}'' = \mathbf{v}' + T$$

$$\mathbf{v}'' = R\mathbf{v} + T$$





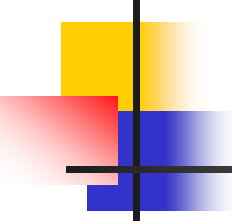
Scaling

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} s_x x \\ s_y y \\ s_z z \end{bmatrix}$$

$$S(s_x, s_y, s_z) = \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & s_z \end{bmatrix}$$

Uniform scaling *iff* $s_x = s_y = s_z$



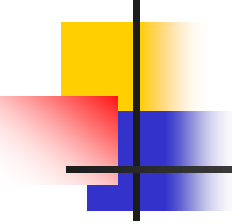


Homogeneous Coordinates

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ can be represented as } \begin{bmatrix} X \\ Y \\ Z \\ w \end{bmatrix}$$

$$\text{where } x = \frac{X}{w}, \quad y = \frac{Y}{w}, \quad z = \frac{Z}{w}$$

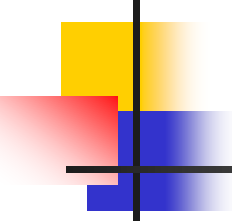




Translation Revisited

$$T(t_x, t_y, t_z) \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & t_x \\ 0 & 1 & 0 & t_y \\ 0 & 0 & 1 & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$



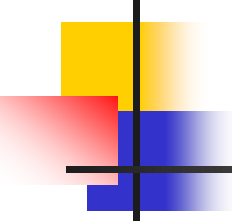


Rotation & Scaling Revisited

$$R_x(\theta) \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta & 0 \\ 0 & \sin \theta & \cos \theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

$$S(s_x, s_y, s_z) \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} s_x & 0 & 0 & 0 \\ 0 & s_y & 0 & 0 \\ 0 & 0 & s_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$





Combining Transformations

$$\mathbf{v}' = S\mathbf{v}$$

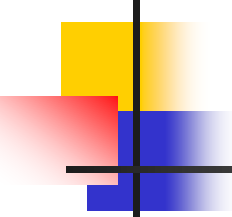
$$\mathbf{v}'' = R\mathbf{v}' = RS\mathbf{v}$$

$$\mathbf{v}''' = T\mathbf{v}'' = TR\mathbf{v}' = TRS\mathbf{v}$$

$$\mathbf{v}''' = M\mathbf{v}$$

$$\text{where } M = TRS$$





Rotations using Quaternions

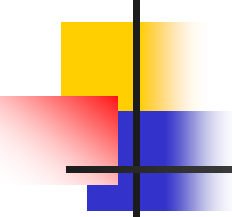
Quaternions

$$q = q_1 + i q_2 + j q_3 + k q_4$$

$$\begin{aligned} i^2 &= j^2 = k^2 = -1 \\ i j k &= -1 \end{aligned}$$

$$q = \begin{pmatrix} q_1 \\ q_2 \\ q_3 \\ q_4 \end{pmatrix} = \begin{pmatrix} s \\ v \end{pmatrix}$$





Rotations using Quaternions

Quaternion multiplication

$$q \ p = \begin{pmatrix} s_q s_p - \mathbf{v}_q \cdot \mathbf{v}_p \\ s_q \mathbf{v}_p + s_p \mathbf{v}_q + \mathbf{v}_q \times \mathbf{v}_p \end{pmatrix}$$

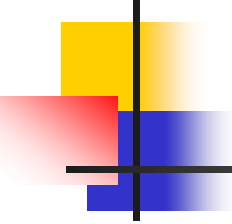
$$[s_1, \mathbf{v}_1] + [s_2, \mathbf{v}_2] = [s_1 + s_2, \mathbf{v}_1 + \mathbf{v}_2]$$

$$[s, \mathbf{v}] \cdot [1, 0, 0, 0] = [s, \mathbf{v}]$$

$$(p \ q) \ r = p \ (q \ r) \quad \text{Associative}$$

$$p \ q \neq q \ p \quad \text{Not Commutative}$$





Rotations using Quaternions

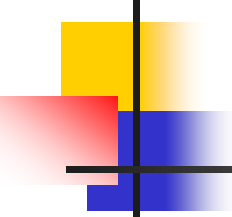
Quaternion magnitude

$$|q| = \sqrt{q_1^2 + q_2^2 + q_3^2 + q_4^2} = \sqrt{s_q^2 + v_q \cdot v_q}$$

Quaternion inverse

$$q^{-1} = \frac{1}{|q|^2} \begin{pmatrix} s \\ -v \end{pmatrix}$$





Rotations using Quaternions

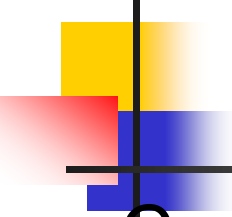
Quaternion rotation:

$$q = \begin{pmatrix} \cos(\theta / 2) \\ \sin(\theta / 2) r \end{pmatrix}$$

$$a = (x, y, z)^T$$

$$rot \begin{pmatrix} 0 \\ a \end{pmatrix} = q \begin{pmatrix} 0 \\ a \end{pmatrix} q^{-1}$$





Rotations using Quaternions

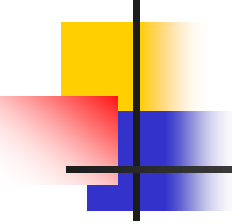
Combining rotations:

$$rot_1 \begin{pmatrix} 0 \\ a \end{pmatrix} = q_1 \begin{pmatrix} 0 \\ a \end{pmatrix} q_1^{-1}$$

$$rot_2 \left(rot_1 \begin{pmatrix} 0 \\ a \end{pmatrix} \right) = q_2 q_1 \begin{pmatrix} 0 \\ a \end{pmatrix} q_1^{-1} q_2^{-1}$$

$$rot_2 \left(rot_1 \begin{pmatrix} 0 \\ a \end{pmatrix} \right) = (q_2 q_1) \begin{pmatrix} 0 \\ a \end{pmatrix} (q_2 q_1)^{-1}$$





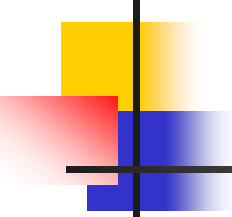
Rotations by Quaternions

$$\text{ROT}_q(\text{ROT}_p(v)) = \text{ROT}_{qp}(v)$$

$$\text{ROT}_{p^{-1}}(\text{ROT}_p(v)) = v$$

$$-q = (-1).(s, v) = (s, v) = q$$





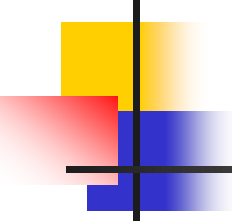
Rotations using Quaternions

Matrix Form:

$$q = w + i x + j y + k z$$

$$\begin{bmatrix} w^2 + x^2 - y^2 - z^2 & 2xy - 2wz & 2xz + 2wy & 0 \\ 2xy + 2wz & w^2 - x^2 + y^2 - z^2 & 2yz - 2wx & 0 \\ 2xz - 2wy & 2yz + 2wx & w^2 - x^2 - y^2 + z^2 & 0 \\ 0 & 0 & 0 & w^2 + x^2 + y^2 + z^2 \end{bmatrix}$$





Helpful Videos

- <https://www.youtube.com/watch?v=d4EgbgTm0Bg>
- <https://www.youtube.com/watch?v=zjMuIxRvygQ>
- <https://www.youtube.com/watch?v=jTgdKoQv738>

